A Resource-Efficient Physics-Based Framework for Real-Time Flood and Landslide Prediction and Damage Assessment

Abstract

Natural disasters such as floods and landslides can cause substantial loss of life, infrastructure, property, and economic resources. Modern disaster prediction systems increasingly employ complex artificial intelligence (AI), machine learning (ML), and deep learning algorithms. Although these methods can provide powerful predictive capabilities, they may require large datasets, substantial computational resources, extensive training, and significant processing time. In time-critical disaster management, computational efficiency and response speed are particularly important because delays in prediction and warning can increase exposure to hazards.

This paper proposes a resource-efficient, physics-based framework for the early prediction of floods and landslides and the subsequent estimation of disaster damage. Instead of relying primarily on complex AI algorithms, the proposed approach begins with the physical causes of each disaster and represents them using established mathematical and physical relationships. For landslides, the framework uses the relationship between driving shear stress and resisting shear strength through the Mohr-Coulomb criterion and Factor of Safety. For flooding, the framework uses a water-balance relationship between incoming water, outgoing water, and the storage capacity of the affected region.

The proposed framework operates in two stages. First, current and forecast environmental conditions are used to estimate how the variables controlling the physical equations will change over time. The predicted variables are then substituted into the corresponding equations to estimate whether the physical failure threshold will be exceeded. Second, historical disaster events are used to establish relationships between the calculated physical hazard indicators and observed damage. Consequently, a new event can be compared with historically observed events having similar physical conditions, allowing an estimate of potential damage.

The framework aims to reduce computational requirements while maintaining physical interpretability, rapid computation, and adaptability. It can potentially provide an efficient foundation for early-warning systems, risk assessment, evacuation planning, and disaster-response decision-making.

Keywords: disaster prediction, flood prediction, landslide prediction, physics-based modeling, Factor of Safety, water balance, damage estimation, early warning system, computational efficiency, disaster management

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1. Introduction

Natural disasters represent a major challenge for disaster-management authorities because their consequences can develop rapidly and affect large populations. Floods and landslides are particularly important because they can be strongly influenced by rainfall, terrain, soil conditions, drainage, groundwater, and other environmental variables.

A disaster-management system has two fundamental objectives:

1. Determine whether hazardous conditions are likely to develop.
2. Estimate the potential consequences if the hazard occurs.

Existing approaches include statistical models, machine learning, deep learning, numerical simulations, and hybrid systems. AI-based methods can identify complex relationships within large datasets, but they may also introduce substantial computational requirements and depend strongly on the quantity and quality of training data.

This paper proposes a different philosophy.

Instead of beginning with the question:

> "Which AI algorithm can predict the disaster?"

the proposed approach begins with:

> "What physical condition actually causes the disaster?"

Once the physical cause is identified, a mathematical representation of that cause can be used as the foundation of the prediction system.

For a landslide, the fundamental question is whether the forces or stresses attempting to move the soil exceed the available resisting strength.

For a flood, the fundamental question is whether water accumulation exceeds the effective capacity of the region to store, absorb, or remove that water.

These relationships provide physically interpretable thresholds that can be evaluated rapidly.

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2. Problem Statement

The central problem addressed in this research is the development of a **fast, computationally efficient, physically interpretable disaster prediction and damage-assessment system** for floods and landslides.

Conventional AI-based prediction systems may require:

- large historical datasets,
- extensive model training,
- high computational resources,
- repeated model inference,
- complex model architectures, and
- substantial preprocessing.

These requirements can become limitations in regions where high-quality disaster datasets are unavailable or computational infrastructure is limited.

Furthermore, a purely data-driven prediction may identify that a disaster is likely without explicitly representing the physical mechanism responsible for the failure.

The proposed research therefore investigates whether established physical relationships can serve as a computationally lightweight foundation for disaster prediction, while historical disaster data can subsequently be used to calibrate damage estimation.

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3. Research Objective

The primary objective is to develop a unified framework capable of:

1. Monitoring current environmental conditions.
2. Forecasting the evolution of important physical variables.
3. Calculating a physically meaningful hazard indicator.
4. Detecting when a physical failure threshold is approached or exceeded.
5. Estimating the potential severity of the event.
6. Comparing the calculated conditions with historical disaster events.
7. Estimating potential damage.
8. Supporting evacuation and disaster-response decisions.
9. Performing these calculations with substantially lower computational requirements than complex AI-based systems.

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4. Proposed Methodology

The proposed framework consists of two principal physical models:

$$\boxed{\text{Flood Model}}$$

and

$$\boxed{\text{Landslide Model}}$$

Both models follow a common principle:

$$\boxed{\text{Failure occurs when environmental demand exceeds system capacity}}$$

However, the physical meaning of demand and capacity differs between disasters.

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5. Physics-Based Landslide Model

5.1 Driving Shear Stress

The proposed landslide model uses the driving shear stress:

$$\boxed{\tau = \gamma z \sin\beta \cos\beta}$$

where:

- γ = unit weight of soil,
- z = depth to the potential failure plane,
- β = slope angle.

The equation represents the shear stress acting along the potential failure plane due to the weight of the soil mass.

5.2 Resisting Shear Strength

The resisting shear strength is represented using the effective-stress form of the Mohr-Coulomb criterion:

$$\boxed{\tau_f = c' + (\gamma z \cos^2\beta - u)\tan\phi'}$$

where:

- c' = effective cohesion,
- ϕ' = effective internal friction angle,
- u = pore-water pressure.

The presence of water is particularly important because increasing pore-water pressure reduces effective normal stress and therefore reduces the available shear strength.

5.3 Factor of Safety

The relationship between resistance and driving stress is represented by:

$$\boxed{FS = \frac{\tau_f}{\tau}}$$

The simplified stability interpretation is:

$$FS > 1$$

indicates that resisting shear strength exceeds driving shear stress, while:

$$FS < 1$$

indicates that the calculated driving stress exceeds the available resisting strength.

Therefore, the proposed triggering condition is:

$$\boxed{\tau > \tau_f \iff FS < 1}$$

This provides a physically interpretable threshold for landslide instability.

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6. Incorporating Time and Forecast Data

A major component of the proposed system is that the equations are not evaluated only using the current state.

Instead, the system estimates how the controlling variables will change over a future time interval.

For example, suppose a weather forecast indicates prolonged rainfall associated with a cyclone.

The system can estimate future rainfall and subsequently estimate variables such as:

$$u(t + \Delta t)$$

The future landslide condition can then be calculated as:

$$FS(t + \Delta t) = \frac{c' + [\gamma z \cos^2 \beta - u(t + \Delta t)] \tan \phi'}{\gamma z \sin \beta \cos \beta}$$

This allows the system to estimate whether the slope is moving toward or away from the instability threshold.

The important concept is therefore:

$$\boxed{\text{Current conditions} \rightarrow \text{Forecast conditions} \rightarrow \text{Future physical state} \rightarrow FS \rightarrow \text{Hazard prediction}}$$

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7. Physics-Based Flood Model

Flooding can similarly be represented as a balance between water entering an area, water leaving the area, and the effective capacity of the area to temporarily store or handle water.

The initial water-balance relationship is:

$$\boxed{P_w = V_w - W_{out}}$$

where:

- V_w = incoming water volume,
- W_{out} = water leaving the area,
- P_w = net water accumulation.

For real-time prediction, the model can be represented dynamically as:

$$\boxed{P_w(t + \Delta t) = P_w(t) + V_w(t) - W_{out}(t)}$$

Let C represent the effective water-holding capacity of the region.

The flood threshold is then:

$$\boxed{P_w > C \Rightarrow \text{Flood condition}}$$

The approach toward the threshold can also be monitored using:

$$\boxed{H_F = \frac{P_w}{C}}$$

where H_F represents the normalized flood-loading condition.

When:

$$H_F < 1$$

the accumulated water remains below the specified capacity.

When:

$$H_F \to 1$$

the region approaches its capacity.

When:

$$H_F > 1$$

the modeled accumulated water exceeds the available capacity.

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8. Factors Affecting the Flood Model

The incoming water volume can be influenced by several factors, including:

- rainfall intensity,
- rainfall duration,
- catchment area,
- snowmelt,
- upstream dam releases,
- dam failure,
- storm surge,
- tributary convergence.

Water removal can depend on:

- river-channel capacity,
- drainage capacity,
- soil infiltration,
- existing soil saturation,
- topography,
- slope,
- other drainage pathways.

The effective storage capacity can depend on:

- floodplain availability,
- wetlands,
- soil characteristics,
- vegetation,
- urbanization,
- impervious surfaces,
- local terrain.

At the current development stage, these relationships are not assumed to have completely calibrated functions because sufficient datasets and development time may not yet be available.

Instead, these relationships constitute the planned expansion of the model.

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9. Common Mathematical Structure

Although floods and landslides are different physical phenomena, their prediction can be represented through a common conceptual structure.

For flooding:

$$\boxed{\frac{\text{Water accumulation}}{\text{Water capacity}} = \frac{P_w}{C}}$$

For landslides:

$$\boxed{FS = \frac{\text{Resisting strength}}{\text{Driving stress}}}$$

Thus, both systems monitor the relationship between an environmental demand and the capacity of the physical system to withstand that demand.

This allows the framework to contain separate physics models while maintaining a common disaster-management architecture.

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10. Historical Damage-Based Assessment

Predicting whether a disaster condition exists is only the first part of disaster management.

The second objective is to estimate its consequences.

The proposed framework therefore introduces a historical-event comparison stage.

For each historical landslide event, the system can calculate:

$$\tau$$

$$\tau_f$$

and:

$$FS = \frac{\tau_f}{\tau}$$

The corresponding observed damage can then be recorded.

For example:

Historical Event	Calculated FS	Observed Damage
Event 1	FS_1	D_1
Event 2	FS_2	D_2
Event 3	FS_3	D_3

A new event produces:

$$FS_{new}$$

The system can then identify historical events with comparable physical conditions and use their observed consequences to estimate potential damage.

The same principle can be applied to flooding using:

$$H_F = \frac{P_w}{C}$$

and historical flood observations.

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11. Damage Prediction Concept

The proposed damage model can initially be represented conceptually as:

$$\boxed{D = f(H, E, V)}$$

where:

- D = expected damage,
- H = physical hazard intensity,
- E = exposure,
- V = vulnerability.

At an early prototype stage, historical matching can be used to approximate this relationship.

With sufficient historical data, the relationship can later be developed into a statistically calibrated function.

For example:

$$D_L = f(FS, \text{exposure}, \text{terrain}, \text{infrastructure}, \ldots)$$

for landslides, and:

$$D_F = f(H_F, \text{water depth}, \text{duration}, \text{exposure}, \ldots)$$

for floods.

This creates a pathway from **physics-based hazard prediction to data-supported damage estimation**.

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12. Why Physics-Based Equations Instead of Complex AI?

The proposed approach does not claim that AI is incapable of predicting disasters. Rather, it addresses situations where computational efficiency, interpretability, and limited data are important.

A complex AI model may require thousands or millions of training examples depending on the problem, together with significant computational resources for training and validation.

In contrast, the proposed physical models directly calculate quantities derived from known physical relationships.

For example:

$$FS = \frac{\tau_f}{\tau}$$

can be evaluated directly once its input parameters are available.

Similarly:

$$P_w(t + \Delta t) = P_w(t) + V_w(t) - W_{out}(t)$$

can be updated continuously without requiring a large neural network inference pipeline.

The proposed advantages are therefore:

Computational efficiency

The mathematical equations require relatively small computational operations compared with training and running large AI models.

Low initial data dependency

The physical model can operate with measurements of relevant physical parameters even when a large labeled disaster dataset is unavailable.

Interpretability

Each variable has a physical meaning.

For example, a reduction in FS can be directly associated with increasing pore-water pressure, changing slope conditions, or other model parameters.

Rapid calculation

The equations can potentially be evaluated continuously as new sensor or forecast data become available.

Transferability

A physics-based model is not inherently limited to the exact historical examples contained in a training dataset.

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13. Importance of Computational Speed

Disaster-management systems are time-sensitive.

A prediction system that can calculate a physically meaningful result rapidly can provide additional time for:

- issuing warnings,
- restricting access to hazardous areas,
- evacuating exposed populations,
- preparing emergency services,
- protecting infrastructure,
- planning response routes.

Therefore, the proposed system treats computational efficiency as a core design objective rather than merely an optimization.

The central design principle is:

$$\boxed{\text{Fast computation} + \text{Physical interpretability} + \text{Sufficient predictive accuracy}}$$

However, computational speed alone is not sufficient. A disaster-warning system must also be properly validated to determine whether its predictions are reliable.

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14. Proposed System Architecture

The complete framework can be represented as:

$$\boxed{\text{Real-Time Environmental Data}}$$

$$\downarrow$$

$$\boxed{\text{Forecast Future Conditions}}$$

$$\downarrow$$

$$\boxed{\left\{\text{Flood Physics Model Landslide Physics Model}\right.}$$

$$\downarrow$$

$$\boxed{\text{Hazard Indicator}}$$

$$\downarrow$$

$$\boxed{\text{Historical Event Comparison}}$$

$$\downarrow$$

$$\boxed{\text{Damage Estimation}}$$

$$\downarrow$$

$$\boxed{\text{Risk Map + Warning + Response Planning}}$$

The final stage can incorporate geographical information to identify exposed infrastructure and population and support evacuation planning.

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15. Future Development

The current equations represent the **baseline physical model**. The next stage is to develop calibrated relationships for the individual variables.

For flooding, future work can establish functions such as:

$$V_w = f(\text{rainfall}, \text{catchment}, \text{snowmelt}, \text{dam release}, \ldots)$$

$$W_{out} = f(\text{drainage}, \text{river capacity}, \text{infiltration}, \text{topography}, \ldots)$$

$$C = f(\text{terrain}, \text{floodplain}, \text{soil}, \text{urbanization}, \ldots)$$

For landslides, future work can improve the prediction of variables such as pore-water pressure:

$$u = f(\text{rainfall}, \text{soil}, \text{groundwater}, \text{antecedent moisture}, \ldots)$$

Historical datasets can then be used to calibrate these relationships.

The damage model can also be improved using larger databases containing:

- disaster intensity,
- geographic location,

- affected area,
- infrastructure exposure,
- population exposure,
- economic loss,
- environmental conditions,
- historical physical parameters.

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16. Validation Strategy

A critical part of future research is validation.

Historical disaster events should be divided into development and testing datasets.

For each historical event:

1. Reconstruct the environmental conditions before the event.
2. Calculate the physical hazard indicator.
3. Determine whether the proposed threshold was exceeded.
4. Compare the prediction with the actual occurrence.
5. Calculate the observed damage.
6. Compare predicted damage with actual damage.

For landslides, the model can evaluate:

$$FS(t)$$

before the historical event.

For floods, it can evaluate:

$$\frac{P_w(t)}{C(t)}$$

before the historical event.

Performance can then be evaluated using appropriate metrics such as:

- detection rate,
- false-alarm rate,
- prediction lead time,
- precision,
- recall,
- damage-estimation error,
- computational time.

This validation is necessary before claiming operational accuracy.

17. Limitations

The proposed framework has several limitations.

First, the physical equations are simplified representations of complex natural systems. Real landslides can involve three-dimensional geometry, heterogeneous soil layers, complex groundwater behavior, vegetation, seismic loading, and progressive failure.

Similarly, flooding can involve complex hydraulic flow, rainfall-runoff processes, river interactions, drainage networks, terrain effects, and changing channel conditions.

Second, accurate prediction requires accurate input data.

Third, the damage relationship cannot be derived solely from the physical threshold. Two events with similar physical hazard indicators may cause different amounts of damage because population density, infrastructure quality, preparedness, and vulnerability differ.

Finally, the proposed framework requires historical validation before its damage predictions can be considered reliable.

18. Conclusion

This research proposes a resource-efficient framework for predicting floods and landslides and estimating their potential consequences.

The central concept is to begin with the **physical cause of the disaster rather than immediately applying a complex AI model**.

For landslides, the proposed model compares driving shear stress:

$$\tau = \gamma z \sin\beta \cos\beta$$

with resisting shear strength:

$$\tau_f = c' + (\gamma z \cos^2\beta - u)\tan\phi'$$

through:

$$FS = \frac{\tau_f}{\tau}$$

with the simplified instability condition:

$$FS < 1.$$

For flooding, the proposed model uses water accumulation:

$$P_w(t + \Delta t) = P_w(t) + V_w(t) - W_{out}(t)$$

and compares it with the effective capacity:

$$P_w > C.$$

The major contribution of the proposed framework is not the invention of these underlying physical equations. These equations are established physical relationships. Instead, the proposed contribution is their integration into a **resource-efficient, time-dependent early-warning and damage-assessment architecture**.

The framework further proposes using historical events to connect physically calculated hazard conditions with observed damage. Thus, once a new event is detected, the system can calculate its physical condition and compare it with historically observed events to estimate potential consequences.

The resulting concept can be summarized as:

$$\boxed{\text{Physical Cause} \rightarrow \text{Mathematical Model} \rightarrow \text{Future State Prediction} \rightarrow \text{Hazard Detection} \rightarrow \text{Historical Compariso}}$$

This approach has the potential to provide a lightweight alternative or complementary layer to computationally intensive AI-based disaster-management systems, particularly in environments where computational resources or high-quality labeled datasets are limited.

The ultimate objective is not simply to determine **whether a disaster will occur**, but to provide sufficient warning and information to determine **what may happen, how severe it may be, and how response resources should be deployed**.